ALICE L WALTON SCHOOL OF MEDICINE

Calcium Channel Blockers

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Learning Objectives (5)

After completing this brick, you will be able to:

1. Explain the mechanism of action and provide a rationale for the clinical indications of CCBs.
2. Classify Calcium Channel Blockers (CCBs) based on their chemical structure.
3. Compare the cardiovascular and hemodynamic effects of CCBs.
4. Explain the clinically relevant pharmacokinetic properties of selected CCB
5. Explain the potential adverse effects, contraindications, and drug interactions associated with CCBs.

CASE CONNECTION

You are a medical student on rotation in the clinic. JD, a 40-year-old male, presents to the clinic. He has been a patient at the clinic for six months. He has a family history of hypertension and has been avoiding visiting the clinic. He does not want to be on any medications. JD's average blood pressure for the last six visits is 155/95 mmHg. During this visit, it was 155/90 mmHg. His heart rate is 80 beats/minute. On the current visit, JD says that he has been experiencing occasional headaches and dizziness.

The doctor advises JD to start taking blood pressure medication. The doctor prescribes amlodipine to help manage his hypertension. JD asks you what does amlodipine do? Does he have to be careful about what he eats or drinks?

Consider your answer as you read, and we'll revisit JD at the end of the brick.

GO TO CONCLUSION ↓

The KIFS courses has discussed the central role of Ca^{2+} at a biochemical and physiological level. We know that Ca^{2+} is an important 2nd messenger that controls a myriad of cellular processes from gene transcription to apoptosis. An increase in intracellular Ca also results in initiating physiological process such as muscle contraction, release of insulin, etc. While Ca^{2+} is stored in the endoplasmic/sarcopla re replenished by Ca^{2+} channels, pumps, and exchangers that line the plasma membrane or are expressed on the ER/SR.

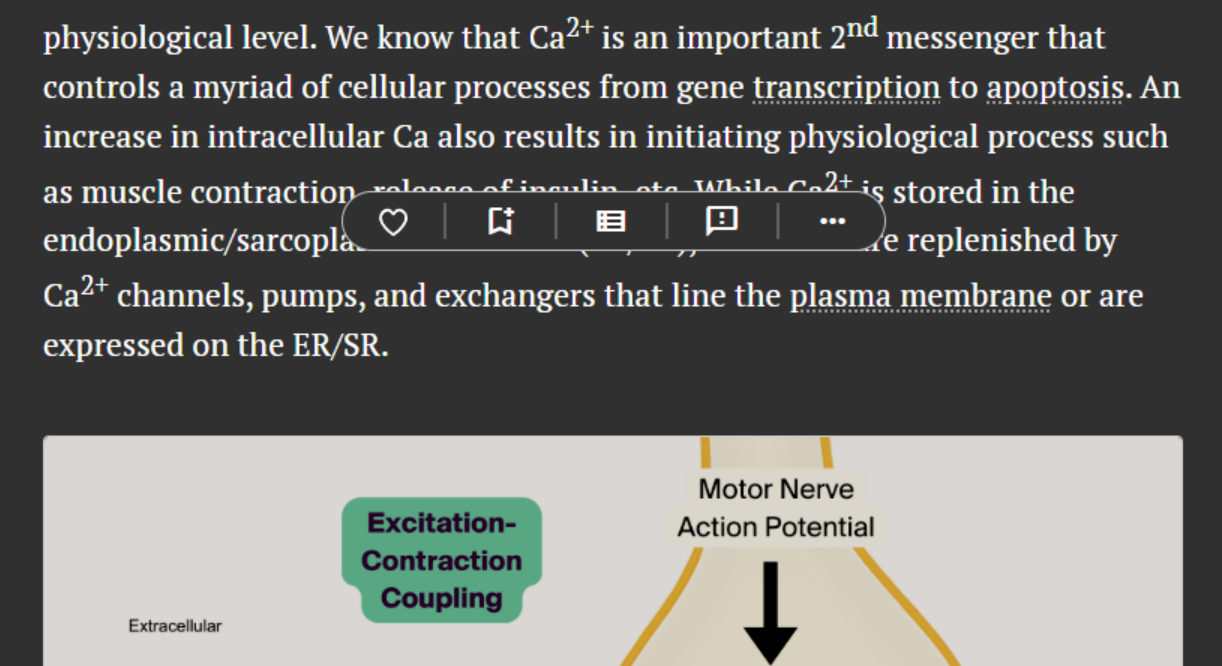
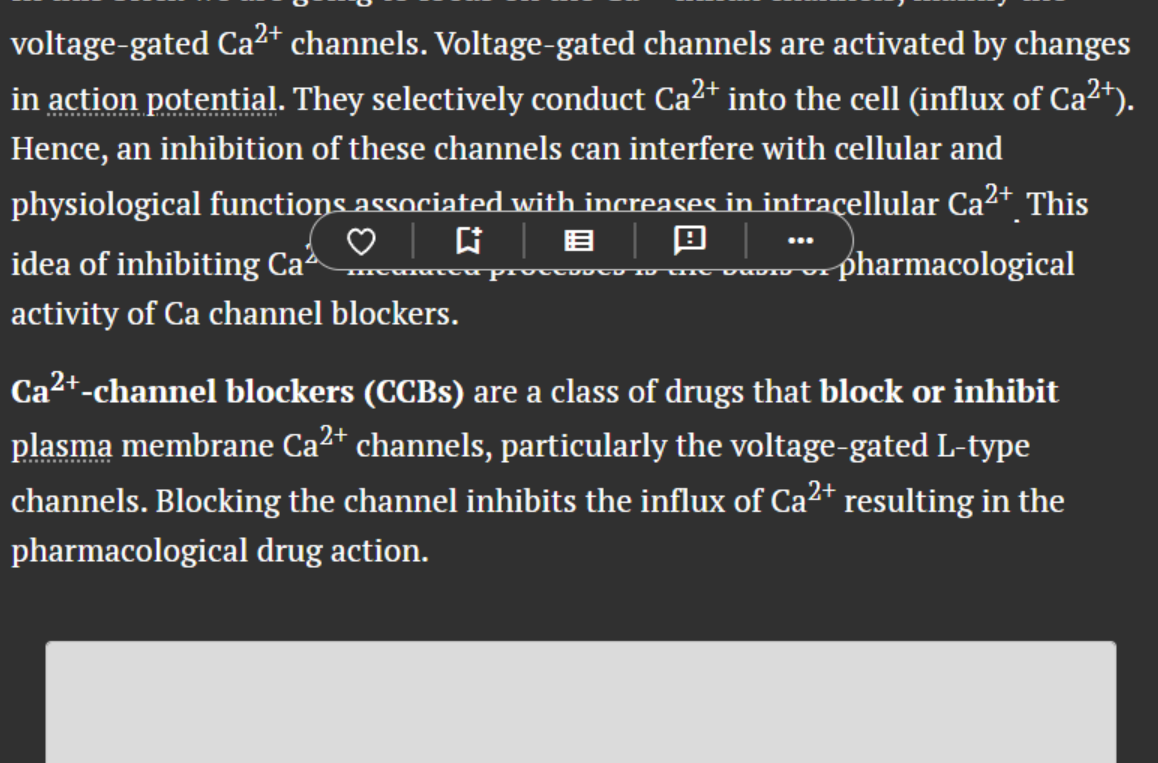

Excitation-Contraction Coupling
Motor Nerve Action Potential
Extracellular
Sarcolemma
T-tubule
L-type Ca2+ Channel
NA+
RYR
CA
Sarcoplasmic Reticulum (SR)
Intracellular

Figure 1

How do calcium channel blockers work? (LO#1)

In this brick we are going to focus on the Ca^{2+} influx channels, mainly the voltage-gated Ca^{2+} channels. Voltage-gated channels are activated by changes in action potential. They selectively conduct Ca^{2+} into the cell (influx of Ca^{2+}). Hence, an inhibition of these channels can interfere with cellular and physiological functions associated with increases in intracellular Ca^{2+}. This idea of inhibiting Ca pharmacological activity of Ca channel blockers.

Ca^{2+}-channel blockers (CCBs) are a class of drugs that **block or inhibit** plasma membrane Ca^{2+} channels, particularly the voltage-gated L-type channels. Blocking the channel inhibits the influx of Ca^{2+} resulting in the pharmacological drug action.


Ca2+
Ca Channel Blockers
L-Type Ca Channel
Ca2+
Influx of Ca initates process such as muscle contraction, insulin secretion, gene transcription

Inhibition of Ca^{2+} by Calcium Channel Blockers

How do we classify calcium channel blockers? (LO#2)

CCBs are classified based on their chemical structure, particularly the presence of a dihydropyridine (Figure 3). Broadly, they are classified as **dihydropyridines (DHPs)** or **non-dihydropyridine (non-DHPs)**. The commonly used CCBs are listed in the table below (Table 1). The names of DHPs end with a suffix "-dipine" (for example Amlo*dipine*). There are two non-DHPs namely verapamil and diltiazem.

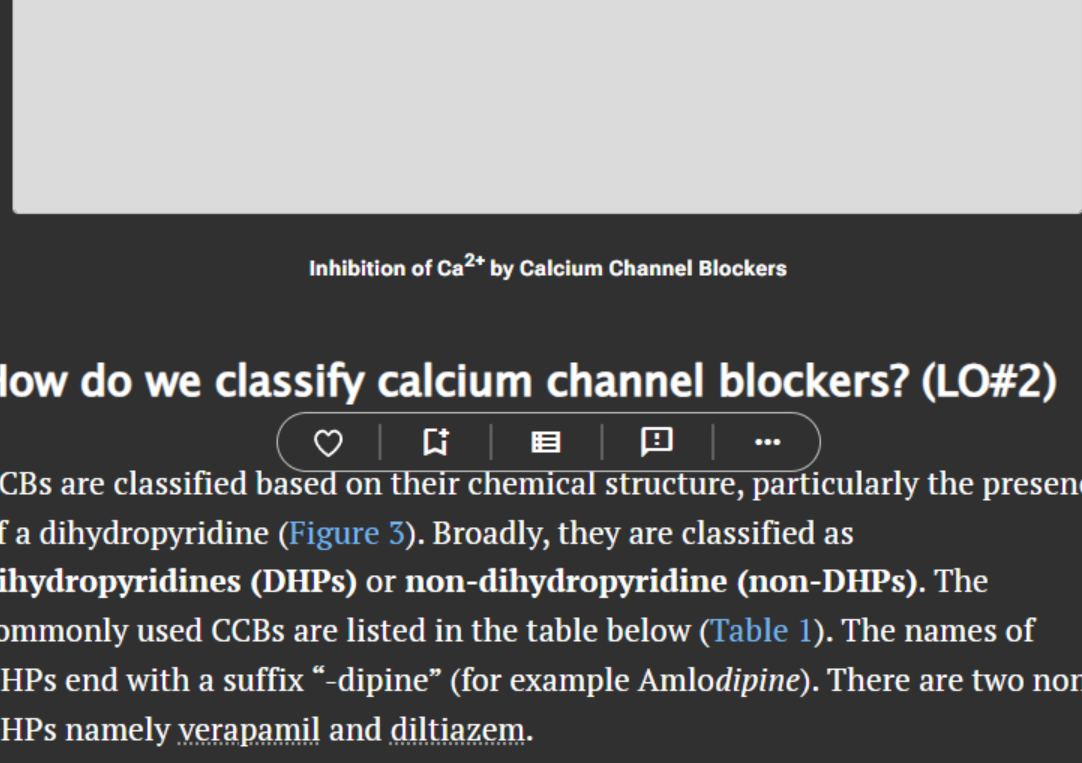

Dihydropyridines (DHPs)
Amlodipine
Nifedipine
Dihydropyridine (DHP) structure
Non-dihydropyridines (Non-DHP)
Diltiazem
Verapamil

Table 1

DHPs	Non-DHPs
Amlodi	apamil
Nifedipine	Diltiazem
NIcardapine	
Felodipine	

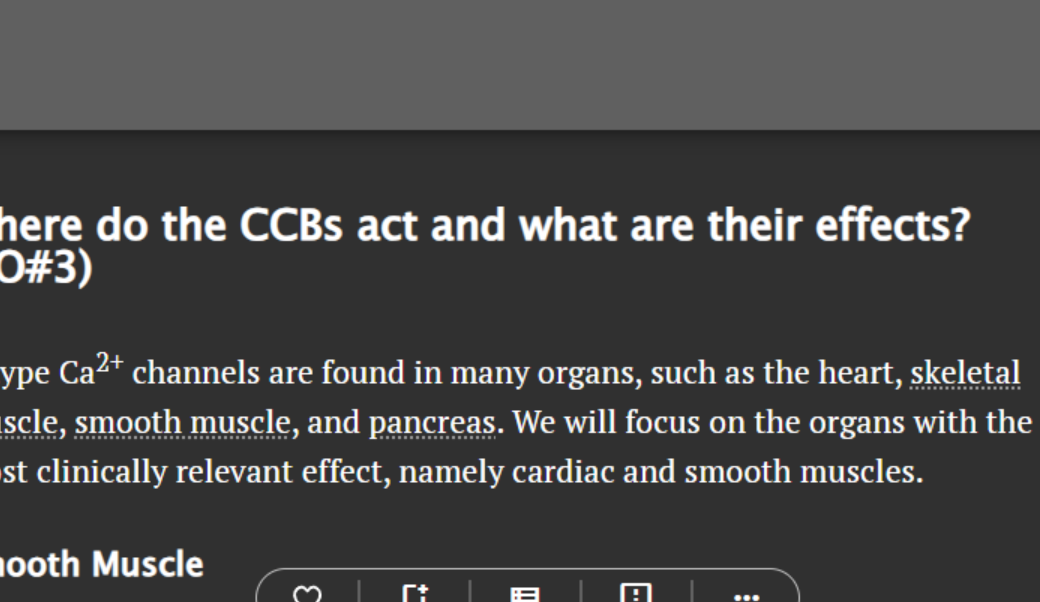
What suffix do dihydropyridine CCBs commonly have?

Where do the CCBs act and what are their effects? (LO#3)

L-type Ca^{2+} channels are found in many organs, such as the heart, skeletal muscle, smooth muscle, and pancreas. We will focus on the organs with the most clinically relevant effect, namely cardiac and smooth muscles.

Smooth Muscle

An increase in cytosolic Ca^{2+} enhances its binding to calmodulin, forming a Ca2+-calmodulin complex. This complex activates myosin light-chain kinase, which phosphorylates the myosin light chain. This phosphorylation facilitates the interaction between actin and myosin, leading to sustained smooth muscle contraction. CCBs inhibit Ca^{2+} influx in the smooth muscles, resulting in a decrease in the contractile process. Thus, vascular smooth muscles relax with CCBs. Arteriolar vascular smooth muscles are more relaxed than the venous muscles. The relaxation of vascular smooth muscles decreases the pressure exerted by the blood on the vascular wall (blood pressure). Hence, these agents are used as anti-hypertensives to treat high blood pressure (Hypertension).


Inhibition by CCBs causes smooth muscle relaxation
Ca2+
Ca2+ Channel Blockers
L-Type Ca Channel
Ca2+
Myosin-light chain
Myosin-light chain PO4
Actin
Smooth Muscle Contraction
Created in https://BioRender.com

How do calcium channel blockers (CCBs) lower blood pressure?

Cardiac Muscle

Cardiac muscles depend on Ca^{2+}-induced Ca^{2+} release (CICR) for initiating each action potential and the sustained contractility of the cardiac muscle. Thus, the inhibition of L-type Ca^{2+} channels in the heart results in slowing down heart rate and the force of contractility. The inhibition of Ca^{2+} influx decreases the rate of SA and AV nodal firing. These agents have negative chronotropic effects. This property can correct abnormal rhythms such as fast heart rates (sinus tachycardia). Hence these agents are used to treat arrhythmias (abnormalities of heart rhythms).

By inhibiting Ca^{2+} influx, CCBs can prevent Ca2+ binding to troponin C, thus inhibiting actin-myosin interactions. This causes a more sustained cardiac relaxation and decreases cardiac muscle contractility. The decrease in cardiac contractility (or negative inotropy) lowers the O2 demand on the heart. In angina, an increased demand for cardiac contraction causes chest pain that can result in ischemia an ic effect of CCBs can decrease the contractility of the heart and are used as antianginal agents.

What is the effect of inhibiting L-type Ca2+ channels in the heart?

CLINICAL CORRELATION

Calcium channel blockers, such as verapamil and diltiazem, exhibit antiarrhythmic properties by primarily targeting the slow response tissues in the SA and AV nodes. These medications are effective in managing supraventricular tachycardias (SVTs) that depend on AV node conduction, such as atrioventricular nodal reentrant tachycardia and atrioventricular reentrant tachycardia. Additionally, they are used to control ventricular rates in conditions like atrial fibrillation, atrial flutter, and atrial tachycardia

Important differences in the selectivity of these agents exist that can affect their cardiac and hemodynamic functions. For example, non-DHPs such as verapamil are potent at SA nodal suppression compared with DHPs. DHPs such as amlodipine are potent arterial vasodilators in comparison to verapamil. Table 2 compares the effects of CCBs (amlodipine is used as a DHP prototype). *This table is very important to understand CCBs clinical indications.* ***This is a must know table !!***

Table 2 Comparison of CCB effects (5 highest to 1 lowest in the ability to inhibit the activity)

Drug	Vasodilation (arterial)	Contractility (Inotropy)	Automaticity (Chronotropy SA nodal activity)
Amlodipine	5	1	1
Verapamil	4	4	5
Diltiazem	3	2	5

Important takeaways from the table

- If you compare Amlodipine (a DHP) to the non-DHPs, you can see that the DHPs are **potent arterial vasodilators.** Hence they are preferred as anti-hypertensives in comparison to non-DHPs.
- Non-DHPs are potent at **suppressing SA nodal activity.** Thus, the are used for their negative chronotropic activity.
- Non-DHPs are potent at **suppressing contractility.** Thus they are used for their ionotropic activity to decrease cardiac contractility.
- Verapamil is potent at SA nodal suppression (negative chronotropic) and is often used as a

Which of the CCB types are more potent vasodilators?

What are important ADME characteristics of CCBs? (LO#4)

Ca^{2+} channel blockers show significant differences in their pharmacokinetics. Immediate-release nifedipine is rapidly absorbed after oral intake with a t1/2 ~1.8 h). This quick absorption can cause a sudden drop in blood pressure, resulting in **reflex tachycardia.** The reflex activation of the sympathetic nervous system, and tachycardia, which may result in flushing.

In contrast, amlodipine is absorbed slowly and has a prolonged effect. With a plasma $t_{1/2}$ of 35 to 50 hours, its plasma levels and effects gradually increase over 7 to 10 days of daily administration at a constant dose, resulting in a steady-state concent . This profile allows the body to adapt and is associated with less reflex tachycardia.

Which CCB types have more potent SA nodal suppression action?

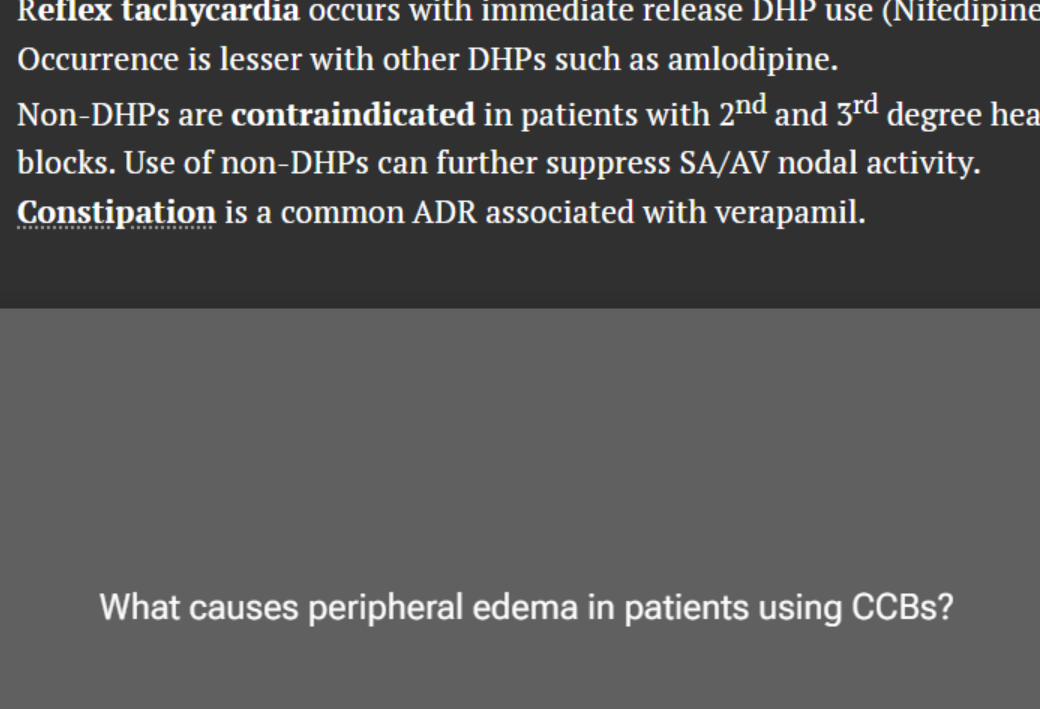
CCBs go through **extensive first-pass.** All CCBs are **substrates of CYP 3A4.** Ca^{2+} channel blockers CYP3A4 inhibitors. Verapamil is particular is a strong CYP3A4 substrate. Verapamil can cause clinically relevant drug interactions with other CYP3A4 substrates, such as *simvastatin* and *atorvastatin.*

On the other hand, the bioavailability of CCBs may be increased by strong CYP3A4 inhibitors, such as macrolides (Azithromycin, clarithromycin) antiretroviral agents, and grapefruit juice. These drug-drug interactions are clinically relevant and should be monitored for patients on these agents.

What are some adverse reactions from CCBs?

- **Peripheral edema:** This is a common ADR associated with CCB use. The edema is not due to fluid retention. It results from the increased hydrostatic pressure in the lower extremities due to capillary dilation of the arterial system. However, venous capillaries are not dilated resulting in the increased hydrostatic pressure and fluid retention.
- Headache, flushing, and palpitations are associated with CCB use particularly after the drug is initiated.
- **Reflex tachycardia** occurs with immediate release DHP use (Nifedipine). Occurrence is lesser with other DHPs such as amlodipine.
- Non-DHPs are **contraindicated** in patients with 2nd and 3rd degree heart blocks. Use of non-DHPs can further suppress SA/AV nodal activity.
- **Constipation** is a common ADR associated with verapamil.

What causes peripheral edema in patients using CCBs?


CCB induced pedal edema : Mechanism
Amlodipine dilates only arterioles (and not the venules)
Increased capillary pressure and capillary permeability
Expulsion of fluid into the surrounding tissue
Pedal edema
Dilates Arterioles
Does not dilate Venules
Dilates Arterioles
Does not dilate venules

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to JD, how do you explain what is the use of amlodipine for JD and what precautions he needs to take with the medication?

JD is prescribed amlodipine, a calcium channel blocker, to help manage his hypertension. Amlodipine works by relaxing the blood vessels, allowing blood to flow more easily and reducing the workload on the heart. You tell JD that this is a commonly prescribed medication for hypertension and is generally safe. However, you tell him to watch for any palpitations and dizziness. He should also check for any feeling of lethargy, which could be a sign of a decrease in heart rate. You ask JD to avoid drinking grapefruit juice. Amlodipine, like other CCBs (DHPs and non-DHPs), is metaboliz inhibitor. You explain to JD that grapefruit juice can interfere with how amlodipine gets metabolized or broken down in the body. This could result in an increased risk of side effects of the drugs.

Summary

- Calcium channel blockers (CCBs) inhibit voltage-gated L-type Ca2+ channels, reducing Ca2+ influx and affecting muscle contraction and heart rate.
- CCBs are classified into dihydropyridines (DHPs) and non-dihydropyridines (non-DHPs), with DHPs having the suffix '-dipine'.
- DHPs primarily cause arterial vasodilation, making them effective antihypertensives, while non-DHPs suppress SA nodal activity and are used for arrhythmias.
- CCBs lower blood pressure by relaxing vascular smooth muscles, reducing vascular resistance, and are used to treat hypertension.
- Non-DHPs decrease cardiac contractility and heart rate, treating conditions like angina and arrhythmias due to their negative inotropic and chronotropic effects.
- CCBs can cause adverse reactions such as peripheral edema, headache, flushing, and reflex tachycardia, with specific interactions due to CYP3A4 metabolism.

Review Questions

1. Which of the follo

A. ○ Amlodipine
B. ○ Nifedipine
C. ○ Felodipine
D. ● Diltiazem

Hide Explanation

The correct answer is Diltiazem (D). Based on its chemical structure, diltiazem is classified as a non-DHP CCB. Amlodipine (A), Nifedipine (B), and Felodipine (C) are DHPs.

2. Which of the following is MOST POTENT negative chronotropic agent?

A. ○ Nifedipine
B. ○ Amlodipine
C. ○ Epinephrine
D. ● Verapamil

Hide Explanation

The correct answer is Verapamil (D). Verapamil is a non-DHP that has potent SA nodal suppression effects. Thus, it acts as a potent negative chronotropic agent. Nifedipine (A) and amlodipine (B) are DHPs that are not as potent negative chronotropic agents. Epinephrine (C) increases the rate of SA nodal firing and, hence, is a positive chronotropic agent.

3. What CYP isoform does verapamil potent inhibit?

A. ○ CYP1A2
B. ○ CYP2C9
C. ○ CYP2C19
D. ● CYP3A4

Hide Explanation

The correct answer is CYP3A4 (D). All CCBs are CYP3A4, however, non-DHPs are potent CYP3A4 inhibitor. Other CYPs such as 1A2 (A), 2C9 (B), 2C19 (C) isoforms are not strongly inhibited by Verapamil.

4. Which CCB is commonly associated with constipation?

A. ○ Amlodipine
B. ○ Nifedipine
C. ● Verapamil
D. ○ Diltiazem

Hide Explanation

The correct answer is verapamil (C). Verapamil is known to cause constipation as a common adverse drug reaction. Amlodipine (A) and Nifedipine (B) are associated with reflex tachycardia and peripheral edema. Diltiazem (D) is a non-DHP and is contraindicated in patients with certain heart blocks.

5. Which CCB is most likely to cause reflex tachycardia?

A. ○ Amlodipine
B. ○ Diltiazem
C. ● Nifedipine
D. ○ Verapamil

Hide Explanation

The correct answer is at can cause vasodilation that can esult in increased sympathetic activity. This is a consequence of short $t_{1/2}$. Amlodipine (A), a DHP, has a longer half-life and can result in less chance of reflex bradycardia. Diltiazem (B) and Verapamil (D) are DHPs that are less likely to cause reflex bradycardia. They have more potent SA nodal suppression.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

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Objective Confidence: 5/5